RhythmNet — Model Card

Model	`munji_ai/models/rhythm_net.py`
Task	Classify 30-second windows of single-lead ambulatory ECG as normal sinus rhythm, atrial fibrillation, or other rhythm
Version	1.0 · October 2026
Repository	`github.com/furatsafa2001/munji_ai`
Author	Safa Khan, AI team
Organisation	مؤسسة عبدالرحمن المعشي لهندسة البرمجيات (Abdulrahman Al-Maashi Software Engineering Est.)

---

0. Where this sits in the client specification

The specification lists twenty conditions across five stages. This model is one stage of five, and covers one of the twenty by name.

Stage	Conditions	Status
1. Signal quality gate	noise, lead-off	written, not trained
2. Rate engine	bradycardia, tachycardia, extreme brady/tachy, pause, asystole, SVT, wide-complex tachycardia	**written and tested**
3a. **Rhythm classifier — this model**	**atrial fibrillation** (atrial flutter folded in)	**trained and evaluated**
3b. Beat classifier	PVC, PAC, couplets, bigeminy, trigeminy	written, not trained
3c. Shockable detector	VF, rapid VT	trained and evaluated
4. ST/T trend	ST elevation, ST depression, T-wave inversion	not built
5. Decision layer	combine, grade, route the alert	not built

What this model is asked for, and what it delivers

The specification names three conditions that a rhythm classifier might carry: atrial fibrillation, atrial flutter, and SVT. It delivers one of them cleanly, and the reasons for the other two are measured rather than assumed.

Atrial fibrillation — delivered. Sensitivity 0.920, specificity 0.973.

Atrial flutter — folded into the AF label, not separate. `windows.py` maps the AFL annotation onto AF rather than OTHER, which is a decision taken before this work and one

the literature supports: the 2025 ConvNeXt detector merges fibrillation and flutter into a single class. It is also the only option the data allows — counting rhythm markers across every registered source gives 59 AFL markers in total. **When this model reports AF, flutter is included in that report.** That must be stated wherever the output is shown.

SVT — not this model's job, by design. It was tried as a fourth class and produced 5 training windows. The reason is arithmetic: a 30-second window needs 21 continuous seconds of one rhythm to carry its label at `RHYTHM_PURITY = 0.7`, and SVT runs are far shorter — a dedicated PSVT study excluded 51 of its own patients for episodes under 10 seconds. SVT is instead handled by the rate engine, where `FAST_NARROW_BPM = 150` and `NARROW_QRS_MS = 120` define it as a rule: fast rate with a narrow complex. Those constants were already in `config.py` before this attempt, which is the clearest statement that the architecture never intended SVT to be a learned class.

The OTHER class carries everything else. It is not a diagnosis. It means "a rhythm that is neither sinus nor fibrillation", and the app must present it that way — as a flag for review, not as a named condition.

---

1. Purpose and scope

Atrial fibrillation is the largest modifiable risk factor for stroke, it is often paroxysmal, and a clinic ECG therefore misses it. Continuous monitoring is the only way to catch it, which is what this model is for.

It does **not** detect ischaemia, infarction, valve disease, heart failure, or conduction block. It does not replace clinical judgement. Its output is a flag for a caregiver, confirmed over about a minute of signal before anyone is contacted.

---

2. Data

2.1 Sources

Database	Version	Records	Native rate	Contribution
CinC Challenge 2017	1.0.0	8,528	300 Hz	the OTHER class, and most of the patients
Icentia11k	1.0	235 patients	250 Hz	ambulatory patch recordings, NSR and AF
LTAFDB	1.0.0	84	128 Hz	long-term AF with episode boundaries
NSRDB	1.0.0	18	128 Hz	screened healthy sinus rhythm
MITDB	1.0.0	48	360 Hz	benchmark, test split only
AFDB	1.0.0	23 usable	250 Hz	benchmark, test split only

Why CinC 2017 was added. Before it, OTHER had 34 training windows against 190 in test — a class the model could not learn but was scored on. Raising Icentia from 145 patients to 235 moved it from 26 to 34, which showed the shortage was in what the sources contain, not how much of them was read: LTAFDB, Icentia and NSRDB are all effectively binary.

CinC 2017 was built for this exact problem. Its task is to sort single-lead recordings into normal, AF, other, or too noisy, and its class counts are N 5,076 / A 758 / **O 2,415** / noisy 279. It also matches this stage precisely: single lead, and every recording is 9,000 samples at 300 Hz — 30.0 seconds, which is the window length. One recording becomes one window.

After adding it, OTHER went from 34 training windows to 2,141.

2.2 Annotation handling

Per-record labels. CinC 2017 holds its labels in a CSV beside the data, one per recording, not in `.atr` files. `baseline_rhythm` could not express that — it is a single rhythm for an entire dataset — so `label_file` and `label_map` were added to the registry and a six-line branch to the loader. The branch fires only for datasets that declare a label file, which is this one and no other.

CSV label	Rhythm token	Class
`N`	`(N`	NSR
`A`	`(AFIB`	AF
`O`	`(OTHER`	OTHER
`~`	`(NOISE`	dropped

Noisy recordings map onto the ambiguous token and are discarded rather than assigned to a class.

Atrial flutter is mapped onto AF upstream, as described in §0.

Validation folder removed before use. CinC 2017 ships a `validation/` directory holding 300 recordings that also appear in `training/` under the same names. Loading both would have fed each of them twice.

2.3 Windows and splits

Parameter	Value
Window	30.0 s (7,500 samples at 250 Hz)
Stride	15.0 s (50% overlap)
Rhythm purity	one rhythm must cover ≥ 70% of the window
Per-patient cap	80 windows
Split	70 / 15 / 15 **by patient**, seed 20260817

Thirty seconds is not arbitrary. Atrial fibrillation is not a shape in one beat; it is an absence — no P wave — and an irregularity with no pattern. Judging that no pattern exists needs enough beats to establish it. At a resting rate a five-second window holds five or six beats, which cannot separate irregular from merely variable. Thirty seconds holds about thirty.

Split	Windows	Patients	NSR	AF	OTHER
train	14,340	5,454	7,656	4,543	2,141
val	3,081	1,163	1,724	871	486
test	6,059	1,206	3,140	2,308	611

Splitting is at patient level, with a leak check at build time.

One caveat on patient identity. CinC 2017 provides no subject identifiers, so each recording is treated as its own patient. If one person contributed two recordings, they could land in different folds. The challenge itself scored recordings independently and nothing in the release allows grouping them, so this is a property of the source rather than a choice.

---

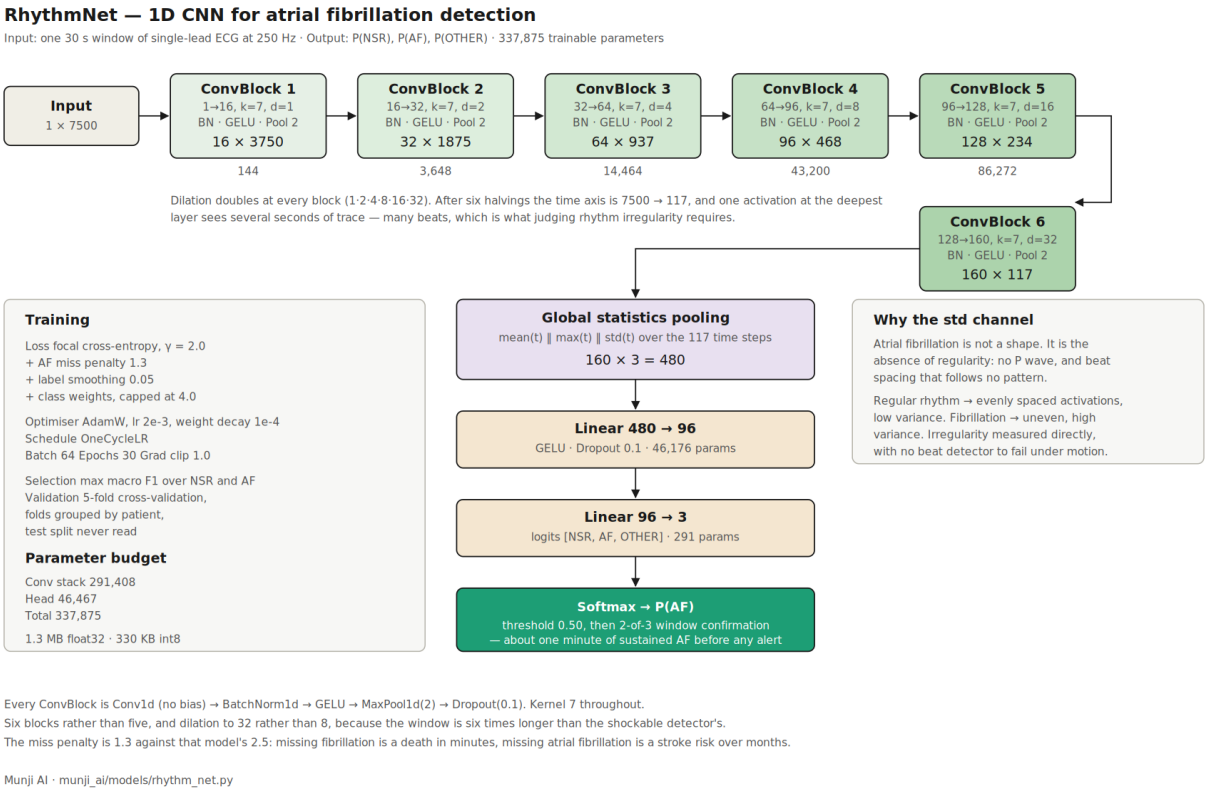
3. Preprocessing

1. **Resample to 250 Hz.** Native rates span 128 to 360 Hz.
2. **Band-pass 0.5 – 40 Hz**, 4th-order Butterworth, zero-phase, so R-peak timing is preserved.
3. **Robust normalisation** per window: subtract the median, divide by the interquartile range. Scale-free, and a single motion spike barely moves it.
4. **Window extraction** as in §2.3.

The signal quality gate runs upstream in deployment at `GATE_MIN_P["rhythm"] = 0.60`, which is deliberately more permissive than the 0.90 the beat classifier demands: rhythm needs R-peak timing, not clean morphology.

---

4. Architecture



4.1 Layer by layer

Stage	Layer	Output	Parameters
Input	—	1 × 7500	—
ConvBlock 1	Conv1d 1→16, k=7, d=1 · BN · GELU · Pool 2 · Drop 0.1	16 × 3750	144
ConvBlock 2	Conv1d 16→32, k=7, d=2 · BN · GELU · Pool 2 · Drop 0.1	32 × 1875	3,648
ConvBlock 3	Conv1d 32→64, k=7, d=4 · BN · GELU · Pool 2 · Drop 0.1	64 × 937	14,464
ConvBlock 4	Conv1d 64→96, k=7, d=8 · BN · GELU · Pool 2 · Drop 0.1	96 × 468	43,200
ConvBlock 5	Conv1d 96→128, k=7, d=16 · BN · GELU · Pool 2 · Drop 0.1	128 × 234	86,272
ConvBlock 6	Conv1d 128→160, k=7, d=32 · BN · GELU · Pool 2 · Drop 0.1	160 × 117	143,680
Global pooling	concat[mean(t), max(t), std(t)]	480	—
Head	Linear 480→96 · GELU · Dropout 0.1	96	46,176
Output	Linear 96→3	3	291
Total			**337,875**

Footprint. 1.32 MB in float32, roughly 330 KB in int8. Six times the input length of the shockable detector, at roughly double the parameters.

4.2 Design rationale

Six blocks, dilation to 32. The input is six times longer than the shockable detector's, so the receptive field has to span many beats rather than one complex. Dilation widens it without adding parameters.

Standard deviation in the pooling. Mean and max are the usual two. The third channel is there because it encodes the distinction this task turns on: regular rhythm produces evenly spaced activations and low variance, fibrillation produces uneven ones and high variance. It measures irregularity without detecting a single beat, which is why it survives motion artifact where an R-peak detector does not.

Raw signal rather than RR-interval statistics. Classical AF detectors measure beat spacing first, then judge its irregularity. This model gets the trace and must find the beats itself. The cost is interpretability; the benefit is that there is no R-peak detector to fail under motion artifact, which is the most common way an ambulatory pipeline breaks. Guided Grad-CAM on a comparable architecture confirmed such networks attend to P-wave absence and R-R irregularity on their own.

A local copy of `ConvBlock`. Each of the four models keeps its own rather than importing a shared one. They are tuned against different failure modes, and a change made for one must not silently alter another.

---

5. Training

5.1 Loss

```
RhythmLoss(
    class_weights   = inverse frequency, capped at 4.0,
    af_miss_penalty = 1.3,
    label_smoothing = 0.05,
    gamma           = 2.0,
)
```

Focal modulation (γ = 2.0) down-weights windows already classified correctly, so the gradient reaches the boundary cases instead of the thousands of obvious sinus windows. It also regularises confidence.

AF miss penalty of 1.3, against 2.5 in the shockable detector. The asymmetry is real but not comparable: a missed fibrillation is a death in minutes; a missed atrial fibrillation is a stroke risk left unflagged over months, with thousands more windows coming from the same patient. Weighting it harder would buy a small gain in recall with a flood of false alarms.

Class weights capped at 4.0. Uncapped inverse frequency gave OTHER a weight near 92 when it held 34 windows. The model would have spent its capacity chasing a class it could not learn. The cap makes a class's weight reflect its importance rather than the scarcity of its examples.

5.2 Optimisation

Optimiser	AdamW, lr 2×10^{-3}, weight decay 10^{-4}
Schedule	OneCycleLR
Batch	64
Epochs	30
Gradient clipping	L2 norm 1.0
Device	Apple MPS

5.3 Model selection

Validation macro F1 over **NSR and AF only**. OTHER is scored, printed and counted in the confusion matrix, but does not choose the checkpoint.

That rule was set when OTHER had 34 windows and its F1 swung on a handful of predictions. With 2,141 windows the case for excluding it is weaker, and it should be revisited at the next training run. It is recorded here because it affects which epoch was kept.

5.4 Validation protocol

Five-fold cross-validation, folds grouped by patient, over the pooled train and validation splits: **17,421 windows from 6,617 patients**. Each patient is predicted exactly once by a model that never saw them. The test split has not been read.

Two seeds were run. Macro F1 came out identical to three decimal places (0.925 and 0.925), and no metric moved by more than 0.005. With 6,617 patients the seed-to-seed variation is negligible, unlike the shockable detector at 199 patients where it reached ±0.012.

5.5 Confirmation rule

`CONFIRMATION["af"] = (2, 3)` : two positive windows among any three consecutive before an alert. At a 15-second stride that is about a minute of sustained AF — deliberate, since paroxysmal AF matters over months and a single noisy window must never raise an alarm.

---

6. Results

Out-of-fold: every window scored by a model that did not train on its patient.

6.1 Headline

Metric	5-fold mean ± sd	Pooled OOF
AF sensitivity	0.926 ± 0.033	0.920
AF specificity	0.970 ± 0.009	0.973
Macro F1 (NSR, AF)	0.923 ± 0.007	0.925
Macro F1 (all three)	—	**0.859**
ROC AUC (AF)	0.982 ± 0.007	0.979
Average precision	—	0.968

Two macro F1 figures appear because they measure different things. The selection metric covers NSR and AF, which is what chose the epoch. The three-class figure includes OTHER and is the one to quote as overall performance.

6.2 Per class

Class	Sensitivity	Specificity	Precision	F1	Support
NSR	0.934	0.901	0.921	0.928	1,702
AF	0.956	0.979	0.949	0.952	899
OTHER	0.710	0.958	0.758	0.733	480

(single-split validation figures; the cross-validated confusion matrix is in §6.3)

6.3 Pooled confusion matrix, 6,617 patients

true ↓ / predicted →	NSR	AF	OTHER
NSR	8,793	220	367
AF	234	5,038	142
OTHER	740	134	1,753

The OTHER row is where the weakness sits: 740 windows of abnormal rhythm were called normal, against 134 called AF. That ratio — confusion with normal far more than with AF — is exactly what the CinC 2017 entries reported. One wrote that the class "is mistaken for AF and even more for Normal", and that the network "struggles to generalise the Other class, maybe the intra-class variability is too impacting". OTHER is not one rhythm; it is every rhythm that is neither of the other two.

6.4 Episode level, after confirmation

Episodes caught	285
Episodes missed	39
Episode sensitivity	0.880
False alarms	25

This metric needed fixing before it meant anything. It first read 0.417 while window-level AF sensitivity was 0.956. The cause was the data, not the model: the rule needs three consecutive windows, and most records here are shorter than that — CinC 2017 gives one 30-second recording per subject, which is one window, and Icentia segments give about two. Every episode in a record too short to confirm counted as missed no matter how well it was classified.

Records that cannot support the rule are now excluded and the exclusion is reported, which is the published convention: short single-recording sets are scored per recording, long-term sets with marked episode boundaries are scored per episode, and clinical studies that do both report them separately. A deployed patch records continuously, so every real episode spans many windows; the exclusion is an artefact of evaluating on archives.

6.5 Per patient

Of 6,617 patients, 6,333 contributed fewer than five windows and are excluded from the per-patient view — a one-window record scores 0.0 or 1.0 and ranks as noise. They are covered by the window-level figures.

Of the 284 patients with five or more windows, 18 have an error rate above 0.20. All are from Icentia11k and LTAFDB: long recordings from real patients, where a single difficult stretch can dominate.

6.6 Comparison with published work

The CinC 2017 task is the closest published benchmark: same four-way framing, same class definitions.

Study	NSR F1	AF F1	OTHER F1	Mean
Challenge winner (2017)	—	—	—	0.826
Philips (Rubin et al.)	0.90	0.80	0.70	0.800
MCNN-BLSTM (2022)	—	—	—	0.846
RhythmNet	**0.928**	**0.952**	**0.733**	**0.859**

The comparison is indicative, not exact: our validation pool mixes CinC 2017 with Icentia, LTAFDB and NSRDB rather than being CinC alone, and the class balance differs. The

pattern holds regardless — strong on the two named classes, and OTHER in line with Philips at 0.70.

---

7. Known limitations

OTHER is a bucket, not a diagnosis. It holds atrial flutter, SVT, bradycardia, block and ectopy together. At 0.710 sensitivity it is the weakest output and must be presented as "a rhythm to review", never as a named condition.

Atrial flutter is reported as AF. Fifty-nine markers across every open source is not enough to separate it. Wherever the AF output is shown, this has to be said.

SVT is not here. It is the rate engine's, by design and by measurement (§0).

Training AF prevalence is unrealistic. AF is 32% of training windows and near zero in a home monitor. At field prevalence the same specificity produces proportionally more false alarms; the confirmation rule and the quality gate are the mitigations.

CinC 2017 was recorded on a handheld AliveCor device, not a chest patch. Electrode placement and contact differ from the deployment hardware.

No subject grouping in CinC 2017. See §2.3.

Single lead. One projection of the cardiac vector, and not designed for ischaemia.

Not clinical judgement. The output is an alert to a caregiver. Every alert should be acted on by a person.

---

8. Files

Path	Role
`munji_ai/models/rhythm_net.py`	Model, loss, class weights
`munji_ai/train/train_rhythm.py`	Training, episode-level reporting
`munji_ai/train/cross_validate.py`	Patient-grouped k-fold
`munji_ai/eval/plot_eval.py`	Confusion, ROC/PR, score distribution, per patient
`munji_ai/rules/rate_engine.py`	Rate conditions including SVT
`munji_ai/data/registry.py`	Dataset definitions, `label_file` mechanism
`munji_ai/data/loader.py`	Annotation parsing, per-record labels
`munji_ai/config.py`	All thresholds

---

9. References

1. Clifford GD, Liu C, Moody B, et al. AF Classification from a Short Single Lead ECG Recording: the PhysioNet/Computing in Cardiology Challenge 2017. *Comput Cardiol.* 2017. https://physionet.org/content/challenge-2017/1.0.0/
2. Rubin J, Parvaneh S, Rahman A, et al. Densely Connected Convolutional Networks and Signal Quality Analysis to Detect Atrial Fibrillation Using Short Single-Lead ECG Recordings. *Comput Cardiol.* 2017. https://arxiv.org/abs/1710.05817
3. Atrial Fibrillation and Atrial Flutter Detection Using Deep Learning. 2025. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12251829/ — ConvNeXt on Icentia11k and LTAF; Grad-CAM on P-wave absence and RR irregularity; AF and AFL merged.
4. An Effective Atrial Fibrillation Detection from Short Single-Lead ECG Recordings Using MCNN-BLSTM Network. *Algorithms.* 2022. https://doi.org/10.3390/a15120454
5. Mukhoti J, Kulharia V, Sanyal A, et al. Calibrating Deep Neural Networks using Focal Loss. *NeurIPS.* 2020. https://arxiv.org/abs/2002.09437
6. Mousavi S, Afghah F, Razi A, Acharya UR. ECGNET: Learning where to attend for detection of atrial fibrillation with deep visual attention. *arXiv:1812.07422.* https://arxiv.org/pdf/1812.07422
7. A deep learning model based on combining surface and esophageal ECG data for diagnosis of paroxysmal supraventricular tachycardia. 2026. https://pmc.ncbi.nlm.nih.gov/articles/PMC13250415/ — PSVT episode durations.
8. Goldberger AL, Amaral LAN, Glass L, et al. PhysioBank, PhysioToolkit, and PhysioNet. *Circulation.* 2000;101(23):e215–e220.
9. Icentia11k — https://physionet.org/content/icentia11k-continuous-ecg/1.0/
10. LTAFDB — https://physionet.org/content/ltafdb/1.0.0/
11. NSRDB — https://physionet.org/content/nsrdb/1.0.0/
12. MITDB — https://physionet.org/content/mitdb/1.0.0/
13. AFDB — https://physionet.org/content/afdb/1.0.0/